

Specialist Masters Programme

Deduction for Late Submission:**Final Mark:**

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1 Methodology

To address Sonic’s challenge of depicting music market categories, a comprehensive network analysis method was employed using the “Deezer” Dataset for Croatia. This approach was chosen for its ability to reveal the hidden patterns in genre relationship and user behaviour, crucial for a major music album called “Sonic”.

The analysis began with data preparation and loading two datasets, ”HR_edges.csv” and “HR_genres.json”, containing friendship network and genre preferences. A bipartite graph was constructed, connecting users to their preferred genres, to get a view of relationship between customers and music genres.

The analysis then focused on genre similarities, projecting a bipartite graph on genres. This step was important for Sonic to understand how music genres relate to each other based on user preferences. The resulting “genre-genre” network was normalised to provide similarity measures, essential for identifying potential genres.

Next, top ten similar genre pairs were identified and ranked to give Sonic clear insights, in order to position new albums and artists more effectively. To visualize genre similarities, a heatmap was created, offering a comprehensive view of market category relationships.

Eigenvector centrality was calculated to identify the most influential genres, to guide Sonic in resource allocation and marketing strategies. The analysis then examined detecting communities using Louvain Algorithm to reveal clusters of related genres, potentially representing different market segments for Sonic to target.

The last part of the analysis focuses on examining how social connections influence music preferences. To get an insight into how social relationships might affect genre popularity and spread, two separate networks: Social and Non-Social networks were created and compared .

Finally, the distribution of genre overlaps in social ties was visualised, offering Sonic a view of how shared musical interests manifest in friendship networks. Therefore, this information will help Sonic to see clear targets in the market, and correctly position new albums and musicians against consumer preferences.

2 Results

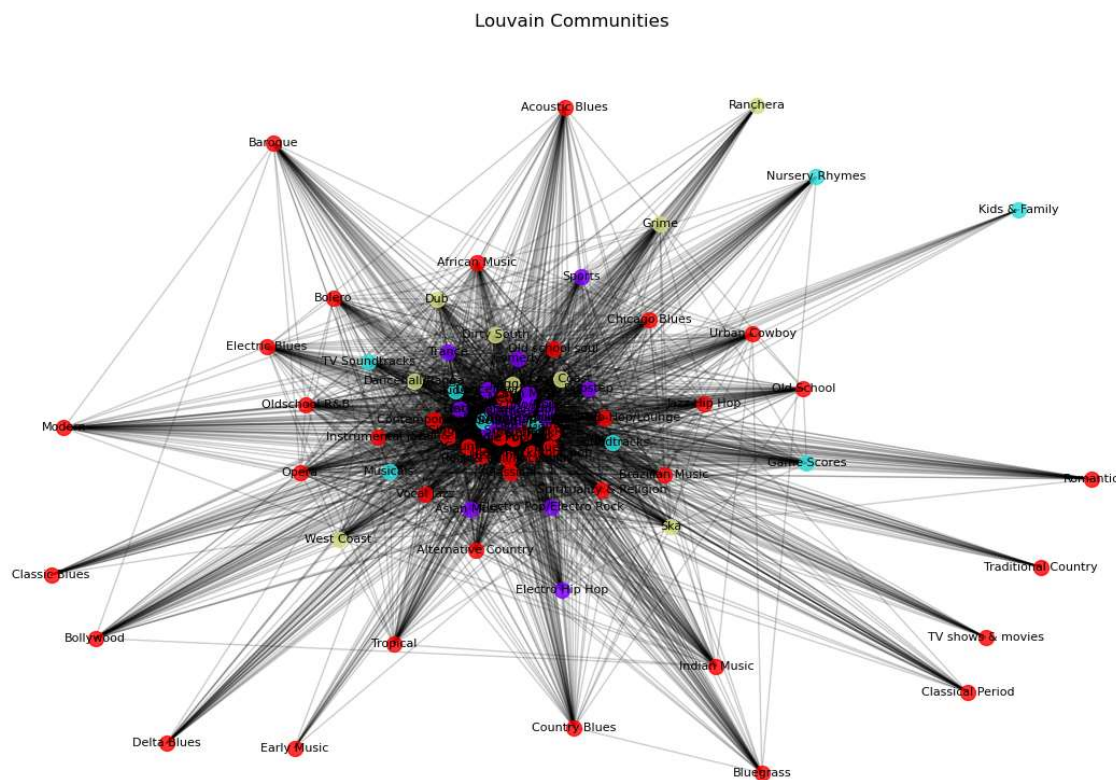
2.1 Genre Similarity Analysis

Similar Pair	Edge Weights
Pop – Dance	0.4508
Pop – Rock	0.4412
Pop – Alternative	0.3601
Pop - Rap/Hip Hop	0.3563
Pop – Electro	0.3302
Dance – Electro	0.3283
Rock – Alternative	0.2917
Rap/Hip Hop – Dance	0.2894
Rock – Dance	0.2890
Pop - R&B	0.2713

Table(i): Top 10 most similar genre pair

In Table(i), “Pop” acts as a dominant connector, appearing in 6 out of 10 strongest pairs, forming the strongest connection with other genres. The strongest relationship exists between “Pop-Dance(0.4508)”, followed by “Pop-Rock(0.4412)”, and last being “Pop-R&B(0.2713)”, indicating these combinations represent the most common cross-genre listening patterns. Here, the edge weights represent normalised co-occurrence frequencies, where higher values indicate stronger relationship between genre in terms of shared listeners and music preference.

2.2 Community Structure and Modularity Analysis



Fig(i): Louvain Communities

Louvain Modularity: 0.0359	
Community Densities	
Louvain Community 1	1.0000
Louvain Community 2	0.8056
Louvain Community 3	0.8611
Louvain Community 4	0.7019

Table(ii): Modularity and community density

The modularity score indicates a relatively weak community structure, suggesting interconnectedness between different music communities.

Louvain Algorithm identified four communities with high internal densities. Community 1, forming central core of the network shows the highest density(1.0000) with the most interconnections. This community includes mainstream genres like Pop, Rock, Dance. Community 2(0.8056), Community

3(0.8611), Community 4(0.7019) forms a distinct cluster while maintaining connections to the main network.

The graph visualization(Fig(i)) shows a very dense structure suggesting strong interconnections between similar genres which is visible through dense edge connections. Some specialized genres like Classical Period, Early Music etc. form smaller subclusters, indicating relatively weaker interconnections.

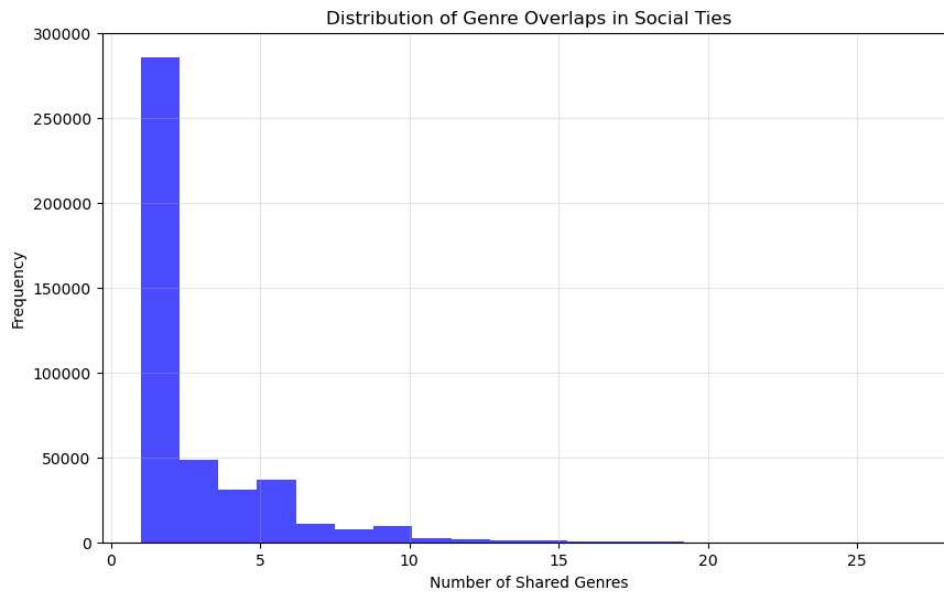
2.3 Social Network Impact

	Social Ties Network	Non-Social Ties Network
Number of edges	439049	59153
Number of nodes	53851	30101
Average clustering coefficient	0.1294	0.0388
Average Degree	16.306067	3.930301
Density	0.000303	0.000131
Components	20	283

Table(iii): Comparing network properties between Social Ties and Non-Social Ties Network

The comparison between social and non-social networks reveals striking differences:

1. The social ties network shows significantly higher connectivity with 439049 edges compared to 59153 in the non-social network. This indicates higher number of nodes represents most users have at least one shared musical interest.
2. Clustering Coefficients of 0.1294 indicates users who share music preferences tend to form more tightly in connected groups. Whereas in non-social, connection without shared music preferences is more random and dispersed.
3. Average degree of 16.306 and 3.93 suggests that each user with shared music tastes connects with about 16 and 4 others on average in social and non-social tie network.



Fig(ii): Distribution of Genre Overlaps

The distribution of genre overlaps in social ties exhibits a highly skewed pattern, with approximately 280,000 connections sharing only 1-2 genres indicating most social connections share only a few genres in common. The frequency drops sharply thereafter, showing a long tail that gradually tapers from 5 to 15 shared genres, with very few connections sharing more than 15 genres.

3 Interpretations

The analysis of Deezer's Music Genre Network and social ties reveals valuable insights for Sonic to see clear targets in the market and correctly position new albums and musicians against customer preference.

The Genre Similarity analysis highlights Pop music as a central hub and acts as a bridge between various genre communities in the network, with strongest ties with Dance(0.4508) and Rock(0.4412). To provide a clear map for strategic artist positioning, the network structure reveals both mainstream clusters and specialized genre peripheries.

The Louvain community detection algorithm identified the 4 Communities with internal densities ranging from 0.7019 to 1.000, suggesting strong genre clusters within the network. With low modularity score(0.0359), shows significant interconnectedness between these communities, suggesting that genre boundaries are fluid and permeable.

The Social Network Analysis demonstrates that the network shows a higher average clustering coefficient(0.1294 versus 0.0388) and average degree(16.31 versus 3.93), indicating that users who are connected through social ties share more similar genre tastes. Furthermore, the histogram of genre overlaps suggests that while most connections share 1-5 genres and a long tail of users sharing up to 25 genres, suggesting the potential for deep musical connections with the social network. This indicates that people tend to form and maintain social ties with others who share similar genres, this phenomenon is known as cultural homophily. This creates a self-reinforcing cycle where friends influence each other's musical tastes over time.

The social network plays a crucial role in shaping genre preferences through two channels:

- Friends actively influence each other's linguistic styles and cultural preferences over time.
- People use music as a way to form impressions of others and establish social connections.

These findings suggest several key insights for Sonic's business strategy. New artists should be positioned within the identified genre communities, leveraging connections between "Pop, Dance, and Rock". The high density of social connections suggests that social network-based marketing strategies and campaigns should be designed to bridge connected genres, allowing for organic growth of artists. The clear community structure also suggests that instead of targeting the entire network, targeting a genre-specific cluster than would be more effective. The distribution of genre overlaps suggests that sonic should focus on creating cross-genre promotional strategies that respect these natural boundaries while facilitating music discovery, which will help Sonic to positing new albums and musicians against customer preference.